Shortage Lists and Monopsony

Applied Labor Economics - Final Report

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1. Context

In this project, we attempt to provide novel evidence on whether a selective, economically-motivated immigration policy generates wage markdowns for vulnerable workers. To do so, we exploit changes in the French *métiers en tension* policy.

On the policy

France mantains lists of occupations facing labor shortages. If an occupation is listed in a given region, employers are exempted from a burdensome standard labour market test when hiring non-EEA nationals (Parlement Français, 2025). The policy aims to lower migrant hiring costs and increase labor inflows into occupations where domestic labour supply falls short of demand. The lists were created in 2008, and have been updated in 2021, 2024 and 2025.

On the monopsony effect

Non-EEA workers employed in a listed occupation may apply for permit extensions or regularization. Leaving a listed job implies losing legal status: *de facto*, it collapses a non-EEA worker's outside option to the process of obtaining a new permit or to undocumented work. French or EEA citizens face no such constraint and retain full mobility.

We presume that firms are aware of non-EEA workers' incentives to be employed in a listed occupation. We posit that firms might exploit this vulnerability to set wages below marginal productivity for this group. We call this the *monopsony effect*, and we attempt to quantify it.

That restricted outside options may translate into wage markdowns is well-documented. In the UAE, tying immigrants to an employer has been found to depress their wages to 51% of marginal product (Naidu et al., 2016). Differences in outside options explain roughly 30% of the immigrant-native wage gap in Germany (Caldwell and Danieli, 2023). Reduced outside options strengthen firms' wage-setting power, particularly at the bottom of the pay distribution (Amior and Stuhler, 2024). An evaluation of the 2008 iteration of the *métiers en tension* policy finds that wages fall more for migrants than for natives in listed occupations, consistently with our framework (Signorelli, 2024).

Other effects

There are competing explanations for the immigrant-native wage gap. An inflow of immigrant workers can heighten labor market competition, affecting peer wages (Albert et al., 2022). Institutional barriers restrict migrants' access to better-paying occupations (Brücker et al., 2018). Employers might also discriminate, which depresses wages through a variety of channels (Becker, 1957; Glover et al., 2017).

Our contribution

We extend Signorelli (2024) by analysing the 2021 and 2025 policy updates, which have not been studied yet, and providing a new theoretical mechanism and framework for analysing this policy.

2. Investigated Mechanism

Wages are the outcome of Nash bargaining between a worker and a firm. A worker with bargaining power $\beta \in (0,1)$, an outside option ω and in a match producing p may expect her wage to be:

$$w^* = (1-\beta)\,\omega + \beta\,p \tag{1}$$

The firm captures $p - w^* = (1-\beta)(p-\omega)$, increasing in the gap between productivity and the outside option.

Natives and EEA citizens can freely switch occupations, their outside option approximates market wage $\bar{w}$. We posit that non-EEA citizens, in particular in listed industries, have a lower outside option $\tilde{\omega} \ll \bar{w}$. Substituting into (1), the predicted markdown is:

$$\underbrace{w^*_{\text{EEA}} - w^*_{\text{non-EEA}}}_{\text{monopsony markdown}} = (1-\beta)(\bar{w} - \tilde{\omega}) > 0 \tag{2}$$

Other forces may shape wages in listed occupations.

- the *shortage premium*, *i.e.* labour scarcity raising wages for all. It is differenced out by comparing coefficients between groups.
- *taste-based discrimination*. If time-invariant, it is removed by our empirical strategy. If time-varying, it is attenuated by restricting the sample to culturally similar workers.
- a *composition effect*, *i.e.* when listing changes the type of worker in a given occupation. We test for it.
- a sudden decrease in *hiring costs* for non-EEA workers increases both their inflow and their wages, assuming that employers do not capture the surplus. Signorelli (2024)'s results dismiss this effect.
- the mentioned *competition effect* is a potential confound. Thus, our estimates should thus be read as an upper bound on the monopsony effect.

3. Empirical Setting

Our analysis relies on 20,172,936 contracts signed from to 2019-2025, associated to 5,113,192 different individuals.

Data sources

Our sample draws from the 2025S2 ForCE vintage. We first use *Fichier Historique* (FH), which records jobseekers registered at France Travail in the past decade. FH provides workers' socio-demographics[1]. We also rely on the *Mouvements de Main d'Œuvre* (MMO), produced by the French Labor Ministry. It contains monthly wages, contract location, and occupation codes (PCS-ESE)[2] for each employment spell.

To understand which contracts were signed under the regulatory regimes of interest, we digitalise the four *arrêtés*, retrieve the targeted occupations codes (FAP[3]) and import them into CASD. We are thus able to assign to each contract a treatment status and a treatment cohort (2008, 2021, 2024, or 2025).

Using ForCE implies everybody in our sample was registered in France Travail, which is no easy feat for non-EEA workers. The represented foreigners will have at the very least a *récépissé*. We expect this selection to bias our effect upward, as we believe these individuals know how to navigate the system better.

A full overview of the data processing is available in Appendix D.

Our sample

As detailed in Appendix D, we restrict *Fichier Historique* to central and south-eastern European workers, in addition to French nationals. The idea is that these workers share relatively similar backgrounds, while differing in permit status[4]. The sample is heavily skewed toward French nationals, as EEA and non-EEA workers each represent under 1.5% of the sample (Table 1).

French workers tend to be younger and more educated, while non-EEA are older and more experienced, yet earn less (Table 1). EEA workers record the shortest employment histories and the lowest diploma rates, while earning the highest average wage. This pattern is likely driven by occupational sorting.

Listed and unlisted occupations differ systematically within each group. Listed workers are older and more likely to be in full-time employment. Generally, within each nationality group, listed occupations pay more at the median than unlisted ones (Appendix A, Figure 2). This is consistent with a *"shortage premium"*, and with listed occupations being concentrated in skilled trades. For non-EEA workers, this listed median advantage is smaller. Non-EEA workers in listed cells earned slightly less than those in unlisted cells in 2021 (€1,364 vs. €1,393), a pattern that reversed in 2025 (€1,537 vs. €1,326) (Table 1 and Appendix A, Table 6).

[1] Especially nationality, which is rarely available.

[2] Professions et Catégories. INSEE classification for survey purposes. Employers will fill out the PCS of their employees on the *Déclaration Sociale Nominative*.

[3] Famille Professionelle. A French Labor Ministry classification, it groups occupations by the type of work actually performed.

[4] Some Bosnians, ethnic Albanians, Turkish may be muslim and face discrimination on religious grounds. In hindsight, a better group to study would have been obtained by comparing Portuguese or Spanish workers with other lusophones or hispanophones.

Table 1: Balance characteristics - 2021 cohort

	French		*EEA*		*Non-EEA*	
	Unlisted	Listed	Unlisted	Listed	Unlisted	Listed
Individuals	4,650,551	368,712	30,234	4,307	55,357	4,031
Contracts	18,261,183	1,534,905	121,424	23,386	214,154	17,884
Monthly wage	1,412	1,629***	1,433	1,615***	1,393	1,364*
Age	31.9	34.0***	35.1	36.2***	37.3	39.2***
Male	0.493	0.576***	0.490	0.583***	0.558	0.517***
Part-time	0.225	0.160***	0.244	0.150***	0.288	0.334***
Experience (months)	47	57***	44	44	56	57
Diploma	0.648	0.668***	0.424	0.265***	0.396	0.372***
Education (NIVFOR)	6.12	5.87***	4.54	3.15***	4.45	3.88***

Stars relate to $\neq$ between listed and unlisted cells within each nationality group ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$).

In 2021, EEA workers were relatively more concentrated in listed occupations (18% of their contracts) relative to French (9%) and non-EEA (10%). By 2025, shares rose across groups, consistently with more occupations being listed (Appendix B, Table 7). Non-EEA see the sharpest increase, with 31% of their contracts in listed cells. This may reflect that the policy is channelling non-EEA hires into shortage sectors, or prior self-selection of non-EEA workers into occupations that were later listed. The second is likely to be true. Indeed, for the 2025 reform, the presence of non-EEA workers in a given occupation×region was itself a listing criterion (Parlement Français, 2025). As we will see, this severely limits the interpretability of our 2025 estimates.

4. Empirical Strategy

Under our hypothesis, listing should raise wages relatively more for French and EEA workers than for non-EEA workers, who face permit-induced disincentives to mobility.

Baseline specification

We estimate separate difference-in-differences regressions for each nationality group $g \in$ {French, EEA, non-EEA}. We study effects around the April 2021 reform (with a ±18-month window) and the May 2025 reform (with a ±6-month window, constrained by data availability)[5].

Identification rests on three simultaneous sources of variation:

- cross-occupation, since not all FAP codes are listed
- cross-region, since listing status varies regionally
- cross-time, before and after each reform

The control group consists of contracts in unlisted occupation×region cells observed over the same window.

The baseline model is the following:

$$\ln w_{irt} = \alpha + \beta\,(\text{listed}_r \times \text{post}_t) + \gamma\,\text{listed}_r + \delta\,\text{post}_t + \mathbf{X}'_{it}\,\theta + \mu_{rt} + \varepsilon_{irt} \tag{3}$$

where w_{irt} is the monthly base wage of worker i in occupation×region cell r at time t, listed_r equals 1 if the cell was first listed in the relevant reform cohort, post_t equals 1 after the reform date. Controls $\mathbf{X}_{it}$ include age, sex, formation, a part-time dummy, and cumulative experience[6].

The coefficient β is the DiD estimator, *i.e.* the wage change in listed cells relative to unlisted cells, after the reform relative to before, for group g. The monopsony effect is recovered by comparing $\hat{\beta}^g$ across groups. The difference $\hat{\beta}^{\text{non-EEA}} - \hat{\beta}^{\text{EEA}}$ cancels the *shortage premium* and isolates the monopsony markdown, under the assumption that non-EEA and EEA workers face similar time-varying discrimination. $\hat{\beta}^{\text{non-EEA}} - \hat{\beta}^{\text{French}}$ is noisier, but a useful comparison for robustness purposes.

On specification choice

We decided to run separate analyses for each reform, and severely limit the analysis window, to avoid using

[5]The 2024 policy update was very limited and is excluded from the analysis.

[6]This control is imperfect. Due to time constraints, we could only use the "experience in sought ROME" variable from France Travail data. Signed contracts might not be in the sought ROME. We deemed it better to use this control than no control at all.

later-listed occupation*region cells as implicit controls for earlier cohorts. This is a documented source of bias in these staggered DiD settings (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). This is why we construct separate "mini-panels", *i.e.* windows of up to 18 months around the studied reform, loosely inspired to what proposed by Cengiz et al. (2019); Baker et al. (2022).

A similar risk is to use already-treated units as controls for later cohorts (**?**). To limit this bias, occupations listed since 2008 are excluded from both groups. Due to a coding mistake, occupation×region cells listed in 2021 and 2024 remained in the 2025 control group while being already treated by then. Fixed effects may absorb their time-varying wage trajectory, but this does not eliminate the bias.

We have also performed Triple DiD regressions. We ultimately decided to exclude it from our analysis, due to its similar results and worse intepretability.

Robustness

Pre-trends and dynamic effects are tested by replacing the "post" dummy by period-by-period interactions.

Each regression is ran using three fixed-effect structures. FE1 (`fap_code + region + year`) controls for time-invariant heterogeneity and common year shocks. FE2 (`fap_code` × `region` × `year`) uses very narrow variation and is a lower bound on the estimates. FE3 (`fap_code` × `year + region` × `year`) absorbs occupation- and region-level time trends.

We test for whether listing changes the type of workers entering the occupation (*"composition effect"*) by re-estimating equation (3) with `formation` as the outcome. A significant $\hat{\beta}$ would indicate a change in the worker profile following the listing, which would bias our estimates.

5. Results

On the validity of assumptions

Event studies show that the parallel trends assumption formally does not hold (Figure 1 and Appendix C, Figure 5).

- For the 2021 study, there is oscillation in the pre-policy period. The most plausible explanation is that the COVID-19 pandemic caused differential shocks between occupations, leading listed-occupation workers to earn relatively less than control workers. This is the case only for French nationals, as immigrants do not exhibit this trend. Our analysis will not retrieve a clean causal effect.

- For 2025, all event studies show a systematic upward slope in the pre-period. This may reflect the endogeneity of the 2025 listing criteria. However, the pre-trend is common across groups, which means it may net out during the comparison. This differential is thus less contaminated than pooled estimates, although it is not causal.

2021 results

If employer bargaining power over non-EEA workers increases in listed sectors, we would expect $\texttt{Treat} \times \texttt{Post}^{\texttt{non-EEA}} < \texttt{Treat} \times \texttt{Post}^{\texttt{EEA}}$. The point estimates go in the opposite direction (Table 2). Both, as well as their differences[7], are insignificant. Results are consistent both in sign and magnitudes across fixed effect structures (Appendix C, Tables 8 and 9).

No significant composition effect is found for any group, suggesting selection bias is unlikely to explain the wage estimates (Table 4).

2025 results

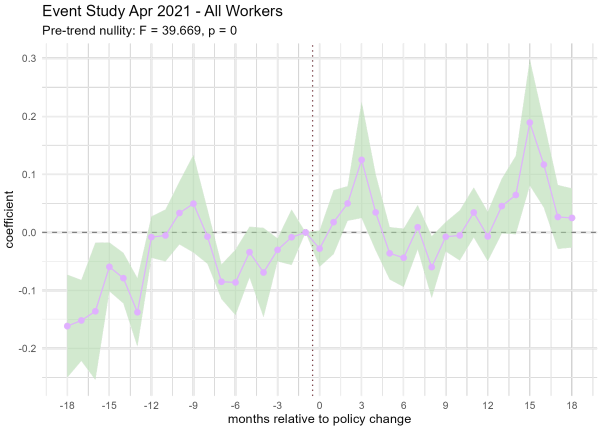
Results are small and insignificant, but the point estimates go in the predicted direction (*i.e.*, non-EEA (+0.012) is below EEA (+0.031)) (Table 3). This is consistent across fixed effect structures (Table 10). The short observation window limits statistical power.

It appears there was a slight upward shift in the formation profile of French workers newly hired in listed occupations. No analogous pattern is detected for non-EEA or EEA workers.

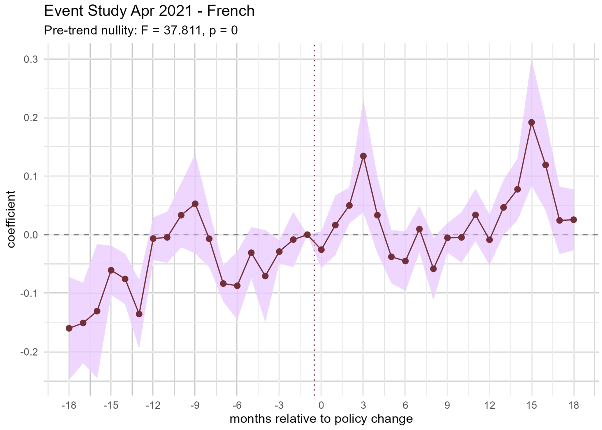
[7]Tested via delta method, not included here.

Figure 1: Event studies by nationality group, 2021 reform

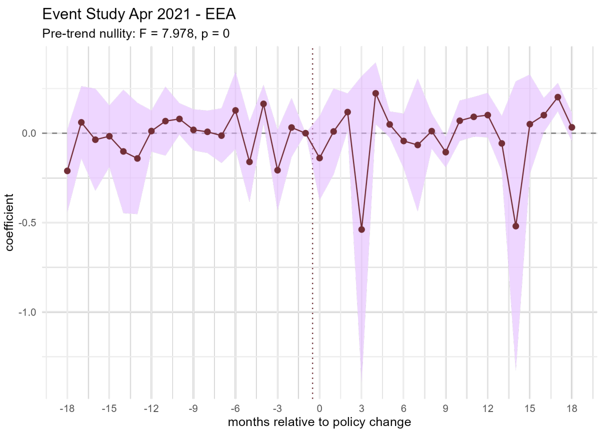
(a) Pooled



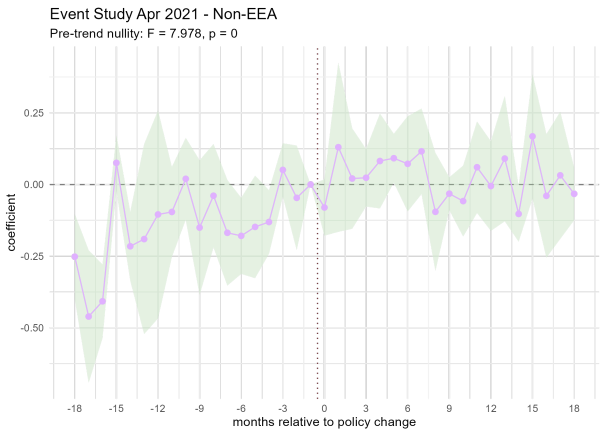
(b) French



(c) EEA



(d) Non-EEA



FE3 specification (`fap_code` × `year` + `region` × `year`), ±18-month window. 95% confidence intervals, SEs clustered at FAP level.

Table 2: 2021 Reform (FE3)

	all	*french*	*EEA*	*non-EEA*
Treatment	0.013	0.013	0.047	-0.048^{*}
	(0.014)	(0.014)	(0.044)	(0.028)
Post	0.019^{*}	0.019^{*}	0.035	-0.004
	(0.011)	(0.011)	(0.028)	(0.023)
Treat × Post	-0.031	-0.031	-0.127	0.051
	(0.020)	(0.019)	(0.081)	(0.043)
Age	0.002^{***}	0.002^{***}	-0.000	-0.000
	(0.000)	(0.000)	(0.001)	(0.001)
Diploma	0.025^{***}	0.025^{***}	0.028^{*}	0.011
	(0.003)	(0.003)	(0.015)	(0.013)
Part-time	-0.712^{***}	-0.710^{***}	-0.669^{***}	-0.833^{***}
	(0.024)	(0.024)	(0.040)	(0.052)
Education	-0.008^{**}	-0.009^{**}	-0.002	0.003^{*}
	(0.004)	(0.004)	(0.003)	(0.002)
Male	0.048^{**}	0.048^{**}	0.067^{***}	0.085^{**}
	(0.023)	(0.023)	(0.023)	(0.034)
Observations	2,284,890	2,241,143	16,173	27,574

Fixed effects: `fap_code × year + region × year`. SEs clustered at FAP level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 3: 2025 Reform (FE3)

	all	*french*	*EEA*	*non-EEA*
Treatment	0.002	0.001	−0.051	0.028^{*}
	(0.015)	(0.015)	(0.040)	(0.017)
Post	0.007	0.007	−0.014	0.025
	(0.011)	(0.010)	(0.020)	(0.021)
Treat × Post	−0.007	−0.008	0.031	0.012
	(0.013)	(0.014)	(0.030)	(0.021)
Age	−0.001	−0.001	−0.001	-0.001^{*}
	(0.000)	(0.000)	(0.001)	(0.001)
Diploma	0.004	0.005	−0.015	0.011
	(0.003)	(0.003)	(0.019)	(0.013)
Part-time	-0.802^{***}	-0.802^{***}	-0.699^{***}	-0.911^{***}
	(0.034)	(0.033)	(0.062)	(0.051)
Education	0.004	0.005	0.007	0.004
	(0.003)	(0.003)	(0.004)	(0.003)
Male	0.064^{**}	0.065^{**}	0.042	0.091^{***}
	(0.027)	(0.028)	(0.028)	(0.033)
Observations	583,337	567,507	6,332	9,498

Fixed effects: `fap_code` × `year + region` × `year`. SEs clustered at FAP level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4: Composition (FE1)

	french	*EEA*	*non-EEA*
Panel A: 2021 reform			
Treatment	0.045^{**}	0.167	-0.241^{*}
	(0.021)	(0.162)	(0.129)
Post	0.005	−0.035	0.090^{*}
	(0.025)	(0.086)	(0.051)
Treat × Post	−0.014	−0.037	0.117
	(0.017)	(0.101)	(0.081)
Observations	2,439,871	17,004	28,835
Panel B: 2025 reform			
Treatment	−0.024	0.021	0.069
	(0.023)	(0.146)	(0.087)
Post	0.001	−0.026	0.098
	(0.015)	(0.127)	(0.080)
Treat × Post	0.044^{*}	0.170	−0.098
	(0.026)	(0.182)	(0.144)
Observations	590,448	6,492	9,692

Fixed effects: `fap_code + region + year`. SEs clustered at FAP level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Note on sample size

Regression sample size drops substantially due to misalignment between the .ods FAP-PCS crosswalk table found on CASD and MMO classifications, generating extensive missings when controlling for FAP. This issue, discovered too late for correction, degrades our estimates and requires resolution in future work.

6. Conclusions

We find no evidence of a monopsony effect. However, data misalignment and design constraints limited our analysis. Future work should correct data cleaning errors, focus on occupations that are similar, and use different samples. Disentangling monopsony from hiring and competition effects requires alternative identification strategies.

Even we found no effect, we like to think our small contribution consists in our framework. Given Europe's demographic decline, labor economics research on migration should shift from assessing native wage impacts to understanding how to attract and retain people under fair and competitive conditions.

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Appendix A - Descriptives

Table 5: Balance characteritics, by nationality group

	French	*EEA*	*Non-EEA*
Individuals	5,019,263	34,541	59,388
Contracts	19,796,088	144,810	232,038
Monthly wage	1,429	1,464***	1,390***
Age	32.0	35.2***	37.4***
Male	0.499	0.502	0.555***
Part-time	0.223	0.234***	0.294***
Experience (months)	48	44***	56***
Diploma	0.649	0.405***	0.395***
Education (NIVFOR)	6.10	4.36***	4.41***

Stars relate to $\neq$ from French ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$).
All EEA vs. non-EEA differences are significant at $p < 0.05$.

Table 6: Balance characteristics - 2025 cohort

	French		*EEA*		*Non-EEA*	
	Unlisted	Listed	Unlisted	Listed	Unlisted	Listed
Individuals	4,409,213	610,050	29,033	5,508	44,706	14,682
Contracts	17,071,372	2,724,716	118,605	26,205	167,625	64,413
Monthly wage	1,423	1,457***	1,435	1,572***	1,326	1,537***
Age	32.0	32.4***	35.0	36.0***	37.1	38.3***
Male	0.493	0.540***	0.497	0.529***	0.534	0.618***
Part-time	0.215	0.280***	0.220	0.304***	0.285	0.307***
Experience (months)	48	46***	44	45	55	61***
Diploma	0.656	0.599***	0.412	0.367***	0.416	0.330***
Education (NIVFOR)	6.20	5.33***	4.44	3.96***	4.62	3.77***

Stars relate to $\neq$ between listed and unlisted cells within each nationality group ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$).

Figure 2: Wage distribution, by nationality group

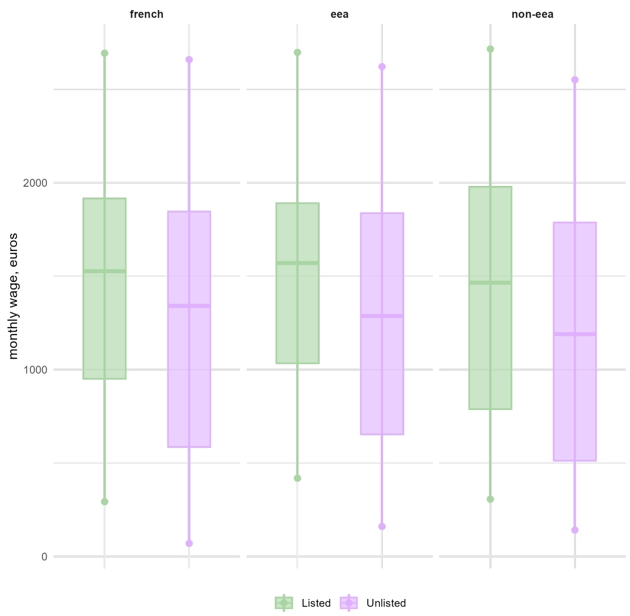
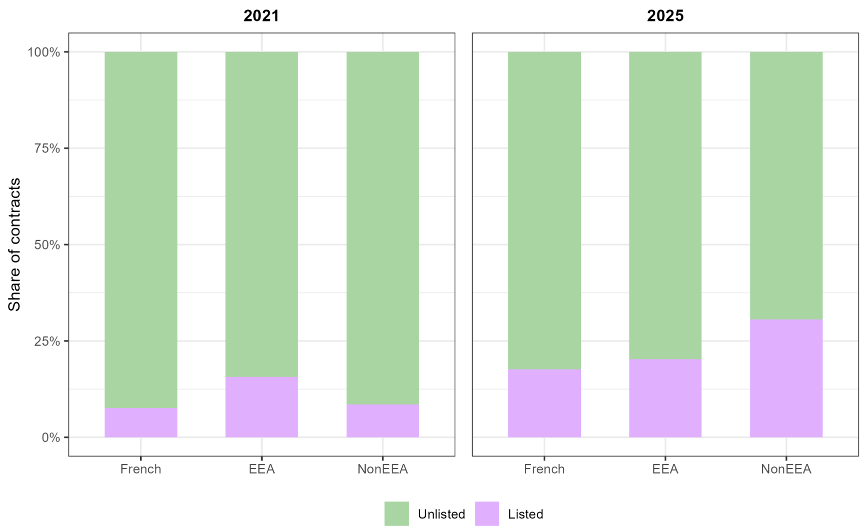


Figure 3: Share of contracts, by nationality group

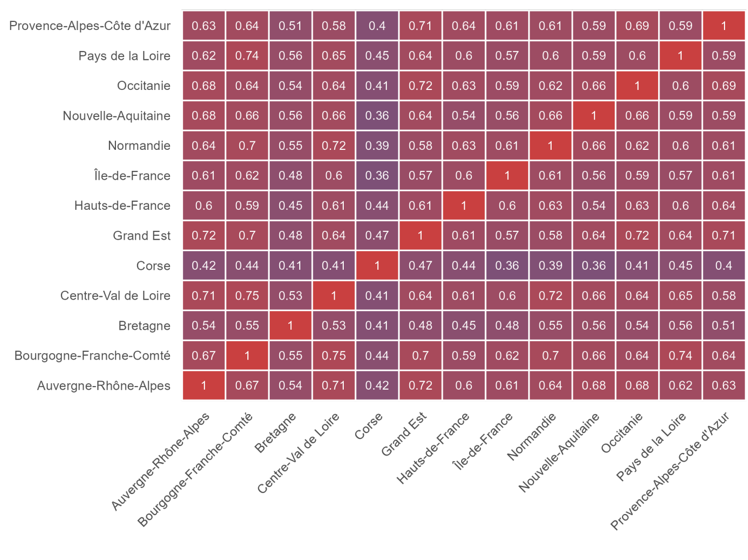


Appendix B - Shortage List

Table 7: Listed FAP, by region and listing year

region	*2021*	*2024*	*2025*	*total*
Auvergne-Rhône-Alpes	21	2	31	74
Nouvelle-Aquitaine	24	2	28	74
Provence-Alpes-Côte d'Azur	18	4	30	71
Bourgogne-Franche-Comté	24	4	22	70
Centre-Val de Loire	26	3	21	70
Grand Est	16	4	25	69
Occitanie	26	2	23	69
Pays de la Loire	26	4	18	69
Hauts-de-France	24	2	21	65
Normandie	23	4	15	64
Île-de-France	20	4	17	63
Bretagne	27	3	14	57
Corse	13	4	20	47

Figure 4: Heatmap of listed occupations



Appendix C - Additional Results

Table 8: 2021 Reform (FE1)

	all	*french*	*EEA*	*non-EEA*
Treatment	0.006	0.006	0.028	-0.047^{*}
	(0.013)	(0.013)	(0.033)	(0.026)
Post	0.019	0.019	0.016	-0.005
	(0.012)	(0.012)	(0.027)	(0.024)
Treat × Post	-0.018^{*}	-0.018^{*}	-0.088	0.040
	(0.010)	(0.009)	(0.072)	(0.032)
Age	0.002^{***}	0.003^{***}	-0.000	-0.000
	(0.000)	(0.000)	(0.001)	(0.001)
Diploma	0.026^{***}	0.026^{***}	0.031^{**}	0.011
	(0.003)	(0.003)	(0.015)	(0.014)
Part-time	-0.713^{***}	-0.711^{***}	-0.673^{***}	-0.834^{***}
	(0.025)	(0.025)	(0.038)	(0.052)
Education	-0.008^{**}	-0.009^{**}	-0.001	0.004^{*}
	(0.004)	(0.004)	(0.003)	(0.002)
Male	0.050^{**}	0.049^{**}	0.068^{***}	0.081^{**}
	(0.023)	(0.023)	(0.024)	(0.033)
Observations	2,284,890	2,241,143	16,173	27,574

Fixed effects: `fap_code + region + year`. SEs clustered at FAP level.

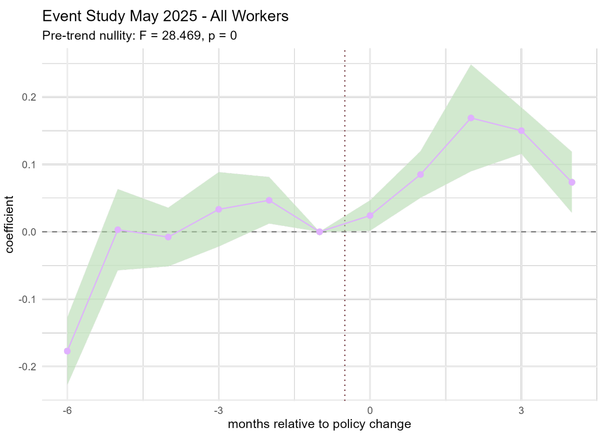
Table 9: 2021 Reform (FE2)

	all	*french*	*EEA*	*non-EEA*
Post	0.022^{*}	0.022^{*}	0.006	0.010
	(0.011)	(0.012)	(0.029)	(0.020)
Treat × Post	-0.060^{**}	-0.062^{**}	-0.008	0.070
	(0.030)	(0.029)	(0.074)	(0.045)
Age	0.002^{***}	0.002^{***}	-0.001	-0.000
	(0.000)	(0.000)	(0.001)	(0.001)
Diploma	0.024^{***}	0.024^{***}	0.024	0.009
	(0.003)	(0.003)	(0.014)	(0.013)
Part-time	-0.711^{***}	-0.710^{***}	-0.669^{***}	-0.801^{***}
	(0.024)	(0.024)	(0.044)	(0.048)
Education	-0.008^{**}	-0.009^{**}	-0.002	0.001
	(0.004)	(0.004)	(0.003)	(0.002)
Male	0.048^{**}	0.048^{**}	0.068^{***}	0.082^{**}
	(0.022)	(0.023)	(0.026)	(0.039)
Observations	2,284,890	2,241,143	16,173	27,574

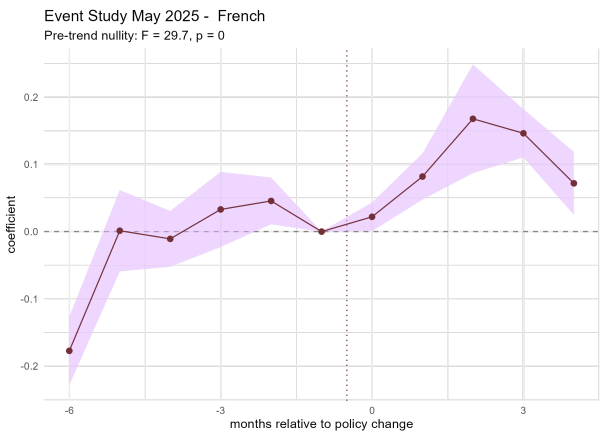
Fixed effects: `fap_code` × `region` × `year` (treatment level absorbed). SEs clustered at FAP level.

Figure 5: Event studies by nationality group, 2025 reform

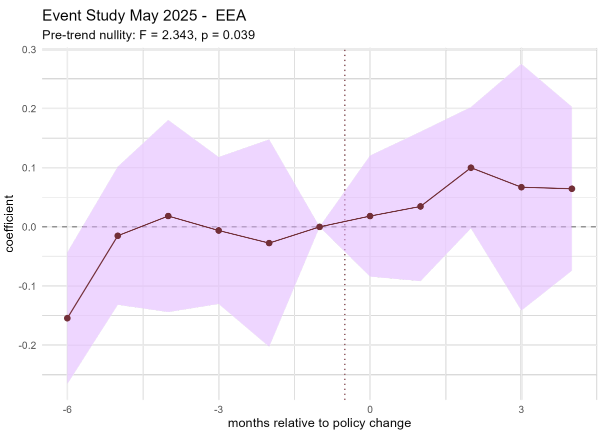
(a) Pooled



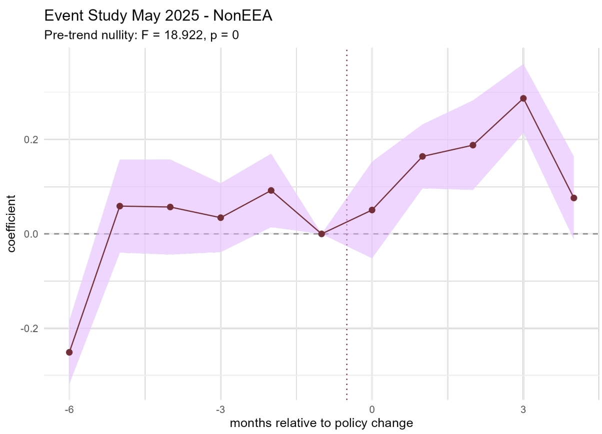
(b) French



(c) EEA



(d) Non-EEA



FE3 specification (`fap_code` × `year` + `region` × `year`), ±6-month window. 95% confidence intervals, SEs clustered at FAP level.

Table 10: 2025 Reform (FE2)

	all	*french*	*EEA*	*non-EEA*
Post	0.006	0.006	−0.007	0.031
	(0.010)	(0.010)	(0.019)	(0.021)
Treat × Post	−0.006	−0.007	0.022	−0.001
	(0.014)	(0.015)	(0.028)	(0.021)
Age	−0.001	−0.001	−0.001	−0.001
	(0.000)	(0.000)	(0.001)	(0.001)
Diploma	0.004	0.004	−0.016	0.018
	(0.003)	(0.003)	(0.020)	(0.013)
Part-time	−0.802***	−0.802***	−0.687***	−0.904***
	(0.033)	(0.033)	(0.069)	(0.057)
Education	0.004	0.004	0.006	0.002
	(0.003)	(0.003)	(0.005)	(0.003)
Male	0.065**	0.065**	0.029	0.079**
	(0.027)	(0.028)	(0.030)	(0.034)
Observations	583,337	567,507	6,332	9,498

Fixed effects: `fap_code` × `region` × `year` (treatment level absorbed). SEs clustered at FAP level

Appendix D - Cleaning Procedure

where	*script*	*input*	*output*
CASD	`01 FH Cleaning.sas`	FH raw	`fh_latest.csv`, `unique_ids.csv`
CASD	`02 MMO Cleaning.sas`	MMO raw	`mmo_latest.csv`
Outside	`01 build shortage list.R`	PDF *arrêtés*, DARES xls	`shortage_list.csv`
CASD	`03 Shortage List + FAP PCS Match.R`	`shortage_list.csv`, PCS-ESE ods	`fap_to_pcs.csv`
CASD	`04 Parquet + Final Cleaning.R`	all CSVs	`mmo_latest_wage.parquet`
CASD	`01 DiD.R`	all parquets	analysis dataset

Part 1: FH Cleaning (`01 FH Cleaning.sas`)

- *Nationality restriction:* retain only spells where `NATION` corresponds to:

Code	Country	Group
01	France	French (reference)
24	Serbia / Bosnia	Non-EEA
26	Kosovo / Montenegro / Macedonia	Non-EEA
29	Turkey	Non-EEA
27	Croatia	EEA
96	Slovenia	EEA
97	Bulgaria	EEA
98	Romania	EEA
19	Greece	EEA
02, 91–95	Baltics, Hungary and Slavic Central Europe	EEA

- *Temporal restriction:* keep spells with `datins` $\geq$ 1 January 2019 to ensure at least two pre-treatment years before the April 2021 reform.
- *Education:* `NIVFOR` alphanumeric codes (`AFS`, `CP4`, ..., `NV1`) are mapped to 0-9 scale (`formation`) for use as a continuous control.
- *One row per worker:* FH records unemployment spells. Multiple rows may exist per `id_force`.
  - Workers with a single spell: kept as-is.
  - Workers with multiple spells: the row with the latest `datins` is retained for the sake of time. Matching each contract to the relevant characteristics would have taken too much time, and given that not many people were involved we decided to accept this error.
- *ID export:* distinct `id_force` values are exported to `unique_ids.csv` and used as a hash reference.

Part 2: MMO Cleaning (`02 MMO Cleaning.sas`)

- *ID filter:* hash join on `unique_ids` restricts MMO to workers present in FH. This is necessary to attach nationality and controls from FH.
- *Contract filters:* applied to each annual file 2019-2025:
  - drop rows with missing `L_Contrat_SQN`, as these contracts have ended and are recorded in other vintages.
  - Drop contracts starting before 2019.
  - Drop contracts shorter than 31 days. Very short spells are either *période d'essai* or do not reflect stable employment.
- *Deduplication across annual files:* the same contract can appear in multiple annual MMO files (once per year until its end date), with later files potentially containing corrections. A contract-level index is built across all years. For each `L_Contrat_SQN`, only the most recent vintage is kept.
- *Policy timing dummies:* binary indicators are created for contract start after each reform date: `post_apr2021` (1 Apr 2021), `post_jan2024` (29 Jan 2024), `post_mar2024` (1 Mar 2024), `post_apr2025` (1 Apr 2025), `post_may2025` (21 May 2025). Symmetric two-year DSN recency windows were also created for robustness checks, but we did not use them.

Part 3. Shortage List Construction and FAP-PCS Matching (`01 build shortage list.R`, `03 Shortage List + FAP PCS Match.R`)

- The four *arrêtés* (2008, 2021, 2024, 2025) are parsed from PDF. Each FAP code is mapped to PCS-2003 codes via a DARES correspondence table. Note that this matching relates only to treated occupations.
- The 2024 amendment is applied as a set of targeted removals and additions to the 2021 list. Delisting and relisting flags are constructed.
- On the CASD, we verify that the PCS codes in `shortage_list.csv` align with the PCS-ESE codes recorded in (`Table_passage_FAP_PCS-ESE.ods`), which is a table we found in the CASD, and in MMO. A mistake occurred here: we did not detect that the .ODS table present on the CASD did not fully align with the MMO.
- The FAP-PCS crosswalk and the treatment status assignment are not merged at this stage.

Part 4. Wage Harmonisation and Geographic Scope (`04 Parquet + Final Cleaning.R`)

- *format conversion:* all CSVs are converted to Parquet. Subsequent operations run via DuckDB without loading files into memory.
- *wage harmonisation.* a standardised `monthly_wage` is constructed from `salaire_base`, `ModeExercice`, and `salaire_base_mois_complet`:
  - *part-time*: `ModeExercice` $\in \{20, 30, 40, 41, 42\}$, or `ModeExercice = 99` with `salaire_base` $<$ 1,498 (2019 SMIC monthly gross).
  - If part-time and `salaire_base_mois_complet = 1`: `monthly_wage = salaire_base` (already full-month equivalent).
  - If full-time and `salaire_base_mois_complet = 0`: `monthly_wage = salaire_base` $\times 30.4375$ / contract days.
  - All other combinations are set to missing and excluded from regressions.
- *Geographic restriction:* contracts are assigned to the 13 metropolitan regions via a postal-code prefix crosswalk. Contracts with `CP` mapping to *Outre-Mer* (prefixes 97-98) or abroad are dropped, as the shortage lists apply only to metropolitan France.

Part 5. Final Merge and Treatment Assignment (`01 DiD.R`)

- MMO and FH are joined on `id_force`.
- The shortage list is joined on `pcs_code` and `region_contract = region`. Treatment indicators are `COALESCE`d to 0 for unmatched contracts, which form the control group.
- Age (`contract_year` $-$ birth year), a male dummy, a diploma dummy, and experience in months are derived at this stage.